

Hosting & Securing an Apache Web Server on Kali Linux

Overview

This project demonstrates how to set up, configure, and secure an Apache web server on Kali Linux. The goal was to host a custom HTML web page accessible from other devices on the same network while managing network configurations, monitoring logs, and implementing firewall rules to block specific IPs without disrupting normal traffic.

Tools & Environment

- **Operating System:** Kali Linux & Windows 11
 - **Web Server:** Apache2
 - **Network Type:** Local Wireless LAN
 - **Server IP:** 192.168.8.113/24
 - **Client Devices:** PCs on same Wi-Fi subnet
 - **Firewall Utility:** iptables
-

Implementation Steps

❑ Install and Enable Apache Web Server

```
sudo apt update && sudo apt install apache2 -y
```

```
sudo systemctl start apache2
```

```
sudo systemctl enable apache2
```

```
sudo systemctl status apache2
```

Apache now runs as a background service and listens on port 80.


```
root@kali:~# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:08:07:e0:c5:0a brd ff:ff:ff:ff:ff:ff
    inet 192.168.8.113/24 brd 192.168.8.255 scope global dynamic noprefixroute eth0
        valid_lft 2792sec preferred_lft 2792sec
    inet6 fe80::e0:27ff:fe0c:50a:1 scope link noprefixroute
        valid_lft forever preferred_lft forever

root@kali:~# sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 3180 packets, 2288K bytes)
pkts bytes target prot opt in out source destination
0 0 DROP all -- * * 192.168.8.100 0.0.0.0/0

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination

Chain OUTPUT (policy ACCEPT 2180 packets, 800K bytes)
pkts bytes target prot opt in out source destination
0 0 DROP all -- * * 0.0.0.0/0 192.168.8.100

root@kali:~#
```

3 Create and Edit Web Page

Your main website files live inside:

`/var/www/html/`

Create or edit your custom page:

`sudo nano /var/www/html/test.html`

Example content:

```
<html>
```

```
  <head><title>My Apache Server</title></head>
```

```
  <body><h2>Welcome to My Kali Linux Apache Server!</h2></body>
```

```
</html>
```

Test locally:

firefox `http://192.168.8.113/test.html`

4 Test Accessibility from Another Device

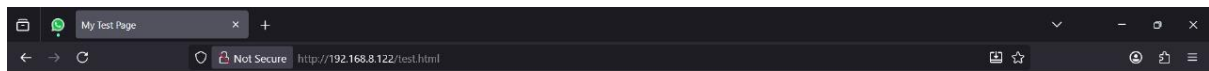
Ensure another PC is on the same Wi-Fi (same 192.168.8.x network).

From that PC:

ping 192.168.8.113

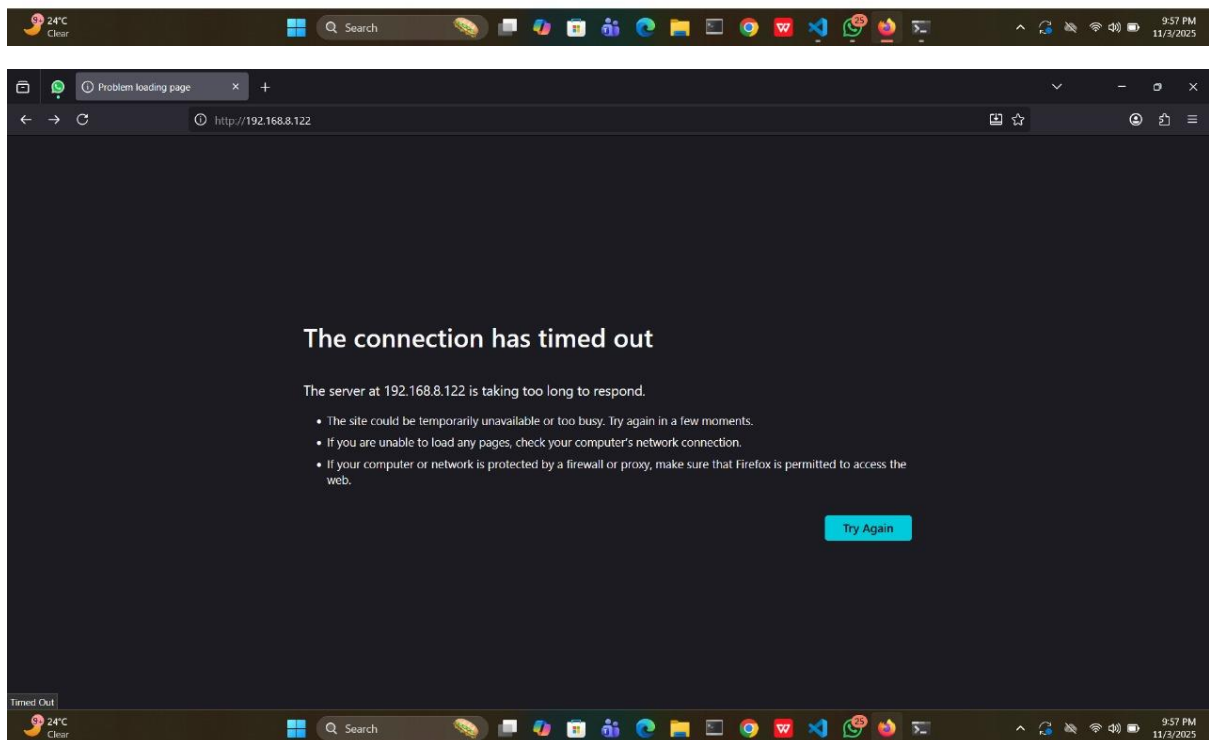
Then open a browser and go to:

<http://192.168.8.113/test.html>



Welcome to my Apache web page!

This page is running on Kali Linux.



50 Configure Firewall to Block a Specific IP

Suppose one client (192.168.8.108) should not access your server.

Block all incoming traffic from that IP:

```
sudo iptables -A INPUT -s 192.168.8.108 -j DROP
```

Block all outgoing traffic to that IP:

```
sudo iptables -A OUTPUT -d 192.168.8.108 -j DROP
```

Verify:

```
sudo iptables -L -v -n
```

You'll see DROP entries for the blocked IP.

```
root@kali: ~
Session Actions Edit View Help

root@kali:~# sudo iptables -A INPUT -s 192.168.8.108 -j DROP
root@kali:~# sudo iptables -A OUTPUT -d 192.168.8.108 -j DROP
root@kali:~# sudo iptables -L -v -n
Chain INPUT (policy ACCEPT 2624 packets, 1886K bytes)
pkts bytes target prot opt in out source destination
0 0 DROP all -- * * 192.168.8.108 0.0.0.0/0

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination

Chain OUTPUT (policy ACCEPT 2182 packets, 788K bytes)
pkts bytes target prot opt in out source destination
0 0 DROP all -- * * 0.0.0.0/0 192.168.8.108
root@kali:~#
```

View Server Logs

Monitor traffic and errors in real time:

```
cat /var/log/apache2/access.log
```

```
cat /var/log/apache2/error.log
```

These logs help track visitors, detect failed requests, and debug issues.

```
root@kali: ~
Session Actions Edit View Help

root@kali:~# cat /var/log/apache2/access.log
192.168.1.146 - [30/Oct/2025:17:53:14 +0530] "GET /test.html HTTP/1.1" 404 492 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.146 - [30/Oct/2025:17:53:15 +0530] "GET /favicon.ico HTTP/1.1" 404 401 "http://192.168.8.108/test.html" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.146 - [30/Oct/2025:18:04:09 +0530] "GET /test.html HTTP/1.1" 200 300 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.146 - [30/Oct/2025:18:06:44 +0530] "GET /test.html HTTP/1.1" 200 493 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:09:48:34 +0530] "GET /test.html HTTP/1.1" 200 437 "-" "curl/8.15.0"
192.168.8.113 - [31/Oct/2025:10:04:26 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:04:26 +0530] "GET /icons/blank.gif HTTP/1.1" 200 431 "http://192.168.8.113/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:04:26 +0530] "GET /icons/test.gif HTTP/1.1" 200 513 "http://192.168.8.113/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:04:26 +0530] "GET /favicon.ico HTTP/1.1" 404 491 "http://192.168.8.113/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:04:30 +0530] "GET /test.html HTTP/1.1" 200 492 "http://192.168.8.113/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:04:30 +0530] "GET /favicon.ico HTTP/1.1" 404 491 "http://192.168.8.113/test.html" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:06:57 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:06:57 +0530] "GET /icons/blank.gif HTTP/1.1" 200 431 "http://192.168.8.113/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.113 - [31/Oct/2025:10:06:57 +0530] "GET /icons/test.gif HTTP/1.1" 200 512 "http://192.168.8.113/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.8.125 - [31/Oct/2025:10:20:07 +0530] "HEAD /test.html HTTP/1.1" 200 251 "-" "curl/8.14.1"
192.168.8.125 - [31/Oct/2025:10:20:41 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.125 - [31/Oct/2025:10:20:41 +0530] "GET /icons/blank.gif HTTP/1.1" 200 431 "http://192.168.8.113/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.125 - [31/Oct/2025:10:20:41 +0530] "GET /icons/test.gif HTTP/1.1" 200 513 "http://192.168.8.113/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.125 - [31/Oct/2025:10:20:41 +0530] "GET /favicon.ico HTTP/1.1" 404 491 "http://192.168.8.113/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:39:53 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:39:53 +0530] "GET /icons/blank.gif HTTP/1.1" 200 431 "http://192.168.8.122/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:39:53 +0530] "GET /favicon.ico HTTP/1.1" 404 491 "http://192.168.8.122/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:39:53 +0530] "GET /test.html HTTP/1.1" 200 492 "http://192.168.8.122/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:43:32 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:43:32 +0530] "GET /icons/blank.gif HTTP/1.1" 200 431 "http://192.168.8.122/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:43:32 +0530] "GET /icons/test.gif HTTP/1.1" 200 513 "http://192.168.8.122/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:48:39 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:57:43 +0530] "GET / HTTP/1.1" 200 697 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
192.168.8.188 - [03/Nov/2025:21:57:43 +0530] "GET /icons/blank.gif HTTP/1.1" 200 431 "http://192.168.8.122/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0"
root@kali:~#
```

```
root@kali:~# cat /var/log/apache2/error.log
[Thu Oct 30 17:45:37.519258 2025] [mpm_prefork:notice] [pid 5172:tid 5172] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Thu Oct 30 17:45:37.519406 2025] [core:notice] [pid 5172:tid 5172] AH00094: Command line: '/usr/sbin/apache2'
[Thu Oct 30 18:23:06.187355 2025] [mpm_prefork:notice] [pid 5172:tid 5172] AH00170: caught SIGTERM, shutting down gracefully
[Thu Oct 30 21:44:33.876310 2025] [mpm_prefork:notice] [pid 725:tid 725] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Thu Oct 30 21:48:33.888432 2025] [core:notice] [pid 725:tid 725] AH00094: Command line: '/usr/sbin/apache2'
[Thu Oct 30 22:08:03.884232 2025] [mpm_prefork:notice] [pid 681:tid 681] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Thu Oct 30 22:08:03.884666 2025] [core:notice] [pid 681:tid 681] AH00094: Command line: '/usr/sbin/apache2'
[Fri Oct 31 09:35:47.413899 2025] [mpm_prefork:notice] [pid 686:tid 686] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Fri Oct 31 09:35:47.413878 2025] [core:notice] [pid 686:tid 686] AH00094: Command line: '/usr/sbin/apache2'
[Fri Oct 31 10:25:37.196079 2025] [mpm_prefork:notice] [pid 686:tid 686] AH00170: caught SIGTERM, shutting down gracefully
[Fri Oct 31 11:35:36.639691 2025] [mpm_prefork:notice] [pid 724:tid 724] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Fri Oct 31 11:35:36.640195 2025] [core:notice] [pid 724:tid 724] AH00094: Command line: '/usr/sbin/apache2'
[Fri Oct 31 14:46:19.479588 2025] [mpm_prefork:notice] [pid 724:tid 724] AH00170: caught SIGTERM, shutting down gracefully
[Fri Oct 31 17:33:57.186300 2025] [mpm_prefork:notice] [pid 724:tid 724] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Fri Oct 31 17:33:57.127731 2025] [core:notice] [pid 724:tid 724] AH00094: Command line: '/usr/sbin/apache2'
[Mon Nov 03 21:38:00.387124 2025] [mpm_prefork:notice] [pid 729:tid 729] AH00163: Apache/2.4.65 (Debian) configured -- resuming normal operations
[Mon Nov 03 21:38:00.392748 2025] [core:notice] [pid 729:tid 729] AH00094: Command line: '/usr/sbin/apache2'
```

Key Learnings

- Hosting and maintaining a web server on a Linux environment.
- Configuring static IPs and ensuring network stability.
- Using iptables to manage inbound and outbound connections.
- Understanding web logs, HTTP traffic flow, and system security.

Outcome

Successfully hosted a custom web page using Apache on Kali Linux, accessible across the local wireless network. Implemented firewall rules to restrict specific IPs while ensuring normal operation for other clients showcasing a real-world balance between functionality and security.

Final Takeaway

This project demonstrates my ability to manage web infrastructure, system configuration, and network security in a Linux environment. It reflects a practical understanding of server deployment, access control, and monitoring, aligning with roles in Cybersecurity, DevOps, or System Engineering.